MLOps

Intro-duction to MLops

Life-cycle of Data science project

Data science project → Requirements Gathering → Data Analyst / Data Scientist

Domain Expertise / Product owner → Requirements Gathering

Business Analyst → Requirements Gathering

Data Analyst / Data Scientist → Discussion

Discussion → Identify the source of data

Discussion → Cloud or 3rd party API's / IOT

Identify the source of data → Data pipe line (ETL)

Cloud or 3rd party API's / IOT → Data pipe line (ETL)

Data pipe line (ETL) → Deployment

Data pipe line (ETL) → Big Data Engineering

Big Data Engineering → data source → mongo / postgre

DVC (Data version control)

Data ingestion (Read the data from source)

↓

Feature engineering (EDA, Handle missing values / outliers / Encording, scalling, etc.)

↓

Feature selection (Correlation, Forward & Backward elemination, univariate selection)

↓

↓

Model Creation & Hyperparameter tuning. (GridSearchCV, Randomized search CV, keras tuner, Bayesian Optimization - Hyperopt Genetics Algorithm, optuna)

↓ Experiment tracking (MLFlow) → Docker / DAGSHUB

↓

Model Deployment (AWS sagemaker)

↓

Model Monitoring (Graphana)

This is to be done for,
{modularity, Github Action, YAML, Pipeline}
ie industry ready.